Forecasting Global CO_2 Emissions Using Statistical and Machine Learning Models

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***Abstract*—This project forecasts total CO_2 emissions across countries using statistical and machine learning models. Guided by instructor feedback, the scope is refined from per-capita equity analysis to a measurable, data-driven forecasting framework. The study uses the World Development Indicators (WDI) dataset (1970–2023), including GDP, population, energy use, renewable energy share, and life expectancy. Countries are categorized into three emission tiers (High/Medium/Low) based on their average total CO_2 emissions (2018–2022) to ensure balanced representation. We compare Naive, Random Forest, and XGBoost baselines at the global panel level using RMSE, MAE, MAPE, and R^2. The framework emphasizes interpretable drivers of emissions and sets up tier-aware modeling as the next step.**

***Index Terms*—CO_2 emissions, forecasting, machine learning, ARIMA, panel modeling, WDI, climate policy**

I. Introduction

A. Background

CO_2 is the major greenhouse gas driving climate change. Predicting total emissions supports mitigation planning, energy transition, and policy evaluation. While per-capita fairness analyses are conceptually valuable, they are less quantifiable for forecasting. This study instead focuses on **total CO_2 emissions**, enabling direct modeling and clearer interpretation.

B. Problem Redefinition and Motivation

After instructor feedback on the initial per-capita framing, we redefined the task to forecast total CO_2 emissions using a unified WDI-based pipeline. This new framing aligns data, features, and models toward measurable, comparative forecasting.

C. Objectives

- Forecast total CO_2 emissions with statistical and ML models (Naive, ARIMA pilot, Random Forest, XGBoost).
- Identify key socioeconomic and energy-related drivers using WDI indicators.
- Compare models via RMSE, MAE, MAPE, and R^2.
- Develop tier-aware modeling to capture heterogeneity between High, Medium, and Low emission countries.

II. Related Work

Previous works, such as those by Raupach et al. (2007) and Chakravarty et al. (2009), focused on fairness and national responsibility in CO_2 emissions, primarily using per-capita measures. Later research, including Li et al. (2019) and Wang et al. (2021), applied ARIMA and SARIMA models for national-level emission forecasting. More recent studies, such as Abas et al. (2023) and Liu et al. (2024), introduced machine learning methods—Random Forest, XGBoost, and hybrid models—showing that ML approaches can outperform traditional time-series models in both accuracy and generalization. Building on these directions, the present study integrates statistical and ML methods using a single data source (WDI) to forecast total CO_2 emissions globally.

III. Dataset

A. Source and Coverage

We use the **World Development Indicators (WDI)** dataset by the World Bank, covering 200+ countries from 1970 to 2023.

B. Variables

Variable	Description
CO_2 total (Mt)	Annual total CO_2 emissions
GDP (constant USD)	Economic activity indicator
Population	Total national population
Energy use (kg oil pc)	Energy consumption per capita
Renewable energy (%)	Share of renewables in total energy use
LifeExp_years	Average life expectancy (years)

TABLE I
Key WDI variables (1970–2023).

C. Country Grouping (Emission Tiers)

Countries are assigned to **High**, **Medium**, or **Low** emission tiers using mean total CO_2 (2018–2022). This quantile-based grouping ensures balanced distribution across regions.

IV. Methodology

A. Pipeline Overview

Data Preparation: merge WDI indicators, harmonize country-year keys, interpolate short gaps, and normalize where necessary.

Feature Engineering: add growth rates (YoY) for GDP, Population, and Energy Use, plus lag/difference of CO_2; retain *Emission_Tier* and *LifeExp_years*.
Modeling: use Naive, Random Forest, and XGBoost for panel data; ARIMA is used only in country-level pilot tests.
Evaluation: RMSE, MAE, MAPE, and R^2 on held-out years (TimeSeriesSplit).
Extension: tier-aware learning and error-by-tier analysis.

B. Planned Experiments (Roadmap)

Phase	Experiment	Output
1	Global panel: Naive, RF, XGB; global EDA	Table of metrics; visual trends
2	Tier-aware models: per-tier training or tier interactions	RMSE/MAE by tier; error plots
3	Feature studies: growth/lag ablations; importance	Feature ranking; policy insights

TABLE II
PLANNED EXPERIMENTS AND DELIVERABLES.

C. Exploratory Data Analysis (EDA)

Global CO_2 totals rise steadily, accelerating after 2000. High emitters dominate absolute emissions; medium and low tiers remain comparatively stable.

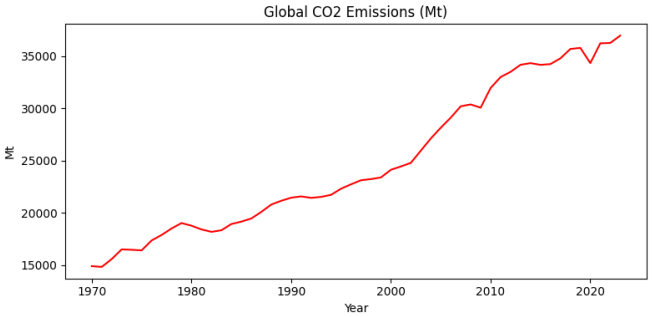

Fig. 1. Global total CO_2 emissions (1970–2023). The trend shows strong growth until the 2010s, followed by mild stabilization.

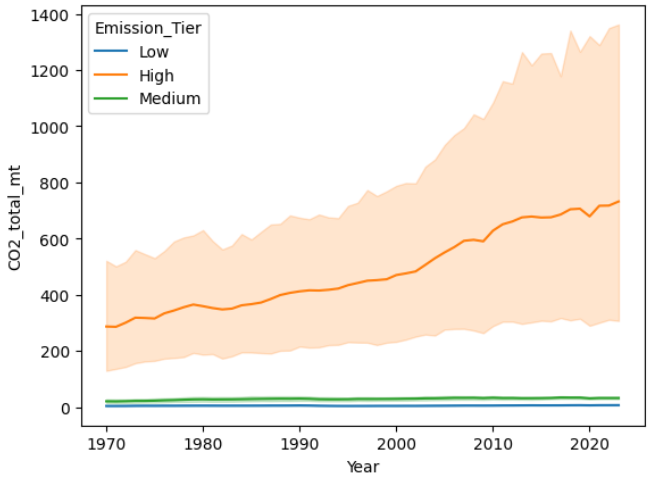

Fig. 2. Average CO_2 trajectories by emission tier (High/Medium/Low). High-tier countries show steep increases; lower tiers are flatter.

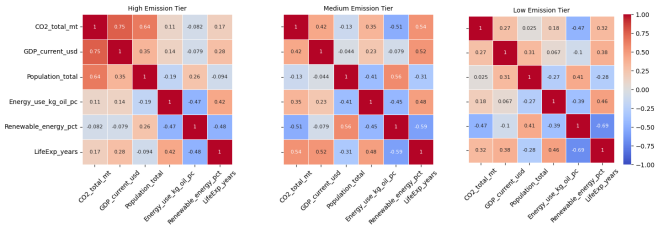

Fig. 3. Correlation heatmap of CO_2 vs key variables (GDP, Energy Use, Renewable Share). GDP and Energy Use correlate strongly with CO_2, while renewables show inverse relation.

D. Modeling Framework

Two complementary approaches are used:

- **Single-country pilots:** ARIMA/ARIMAX models are tested for select countries (China, USA, Thailand) to analyze temporal and exogenous effects.
- **Global panel forecasting:** Combine all countries with encoded *Emission_Tier* features and train Naive, Random Forest, and XGBoost models to capture cross-country structures.

E. Evaluation Metrics

Models are assessed via RMSE, MAE, MAPE (%), and R^2 on held-out test years to quantify predictive accuracy and explanatory power.

F. Preliminary Global Results

Model	RMSE	MAE	MAPE%	R^2
Naive Last Observed	**78.28**	**16.11**	**12.30**	**0.9877**
XGBoost Regressor	142.78	31.34	42.18	0.9625
Random Forest Regressor	154.90	35.00	44.68	0.9463
Mean Baseline	773.33	231.19	3683.12	-0.0041

TABLE III
PRELIMINARY MODEL PERFORMANCE ON GLOBAL FORECASTING.

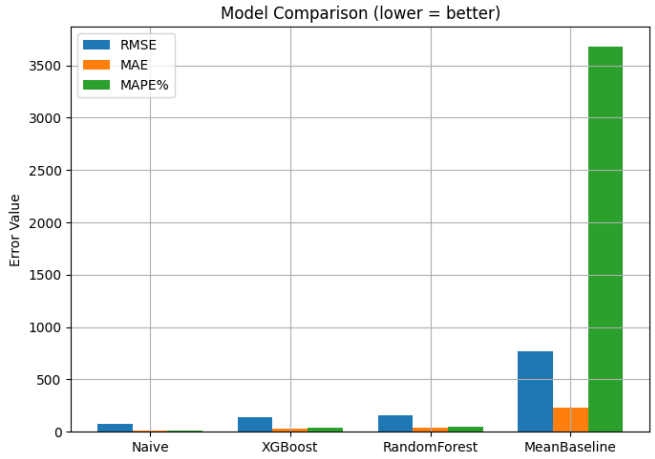

Fig. 4. Comparison of forecasting models (lower = better). Naive dominates due to smooth persistence; ML models show competitive R^2.

G. Discussion and Next Steps

The Naive baseline currently yields the lowest RMSE, showing temporal persistence in emissions. ML models achieve high R^2 (>0.94), demonstrating explanatory strength from WDI variables (GDP, Energy Use, Population, *LifeExp_years*, *Emission_Tier*). Next steps include (i) tier-aware training/evaluation, (ii) adding lag-growth features, and (iii) error analysis by emission tier.

V. Expected Results

- Tier-specific forecasts to 2030 and error breakdowns.
- Enhanced ML accuracy via growth/lag and tier-aware modeling.
- Feature importance showing GDP and Energy Use as key drivers, with renewables inversely linked to emissions.

VI. Conclusion

This study establishes a unified WDI-based pipeline for forecasting total CO_2 emissions using statistical and machine learning methods. By integrating emission-tier grouping and life expectancy as contextual variables, the framework achieves both interpretability and extensibility for future global emission prediction research.

References

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SUMMARY OF FEEDBACK AND RESPONSES

Feedback from Instructor	**Our Response / Adjustment**
Too few samples.	Expanded the dataset to include all available countries from the World Development Indicators (1970–2023), ensuring sufficient samples for each emission tier.
Need to properly define small and large countries.	Replaced this with objective grouping based on total CO_2 emission tiers (high, medium, low emitters) using averages from 2018–2022.
Justify ML model selection.	We adopted a two-level modeling framework. ARIMA and ARIMAX are retained only for single-country diagnostic analysis to capture national time-series dynamics. For the main global forecasting task, we selected Naive, Random Forest, and XGBoost models to represent statistical, ensemble, and gradient-boosting paradigms respectively.
Any correlation of CO_2 emission between countries?	Introduced a global panel model with country-encoded features to capture cross-country temporal relationships.
Main contributions compared to others.	Clarified our key contributions: (1) data-driven total CO_2 forecasting from WDI; (2) growth-based feature engineering; and (3) interpretable feature importance analysis for policy insights.

TABLE IV

SUMMARY OF INSTRUCTOR FEEDBACK AND CORRESPONDING ADJUSTMENTS IN THIS REVISED PROPOSAL.